o6SG

Response: Sensitivity Analysis of WePE Hyperparameters

We thank the reviewer for this targeted question. To provide direct empirical evidence, we conducted a controlled sensitivity analysis sweeping four groups of hyperparameters on a 14×14 patch grid (224×224 images, patch size 16). For each configuration we report three metrics: (i) the Pearson correlation ρ between normalised patch distance and positional-encoding dissimilarity ($1 - \cos\mathrm{sim}$) over all $\binom{196}{2} = 19{,}110$ patch pairs, which quantifies how well the distance-decay property is preserved; (ii) the proxy training loss after 30 gradient steps on a depth-6 ViT-T, which probes training stability; and (iii) feature statistics (mean absolute value, saturation fraction $|f| > 0.99$, and near-zero fraction $|f| < 0.01$).

Table 1: Sensitivity to the pole-neighbourhood multiplier κ.

κ	ρ	Proxy loss	Mean$\lvert f\rvert$	Std	Sat.%	≈0%
1.0	0.6253	4.508	0.1063	0.225	1.91	11.61
5.0	0.6253	4.454	0.1063	0.225	1.91	11.61
15.0	**0.6253**	**4.471**	**0.1063**	**0.225**	**1.91**	**11.61**
30.0	0.6253	4.515	0.1063	0.225	1.91	11.61
60.0	0.6253	4.531	0.1063	0.225	1.91	11.61
120.0	0.6253	4.496	0.1063	0.225	1.91	11.61

(A) Pole-neighbourhood threshold κ ($|z| < \kappa\varepsilon$ triggers $C_{\mathbf{large}}$ substitution). All three metrics are *identical* across the entire range $\kappa \in [1, 120]$ (CV(ρ) = 0.000, CV(loss) = 0.006). The reason is structural: for the lemniscatic square lattice with $\omega_1 \approx 2.622$, the normalised patch coordinates $(u, v) \in (0, 1)^2$ are mapped to $|z| \geq 0.5\omega_1/H \approx 0.094$, which is already $\sim 10^7\varepsilon$ away from the origin ($\varepsilon = 10^{-8}$). Consequently, *no patch coordinate ever falls within the clipping radius for any tested* κ, and the κ parameter is effectively inactive under standard coordinate normalisation. This confirms that Eq. (7) of the paper is robust to the choice of κ over more than two orders of magnitude.

Table 2: Sensitivity to the pole substitute C_{large}.

C_{large}	ρ	Proxy loss	Sat.%
5×10^1	0.6253	4.459	1.91
2×10^2	0.6253	4.500	1.91
5×10^2	0.6253	4.476	1.91
2×10^3	0.6253	4.459	1.91
5×10^3	0.6253	4.488	1.91
1×10^5	0.6253	4.519	1.91

(B) Pole substitute value $C_{\mathbf{large}}$. ρ and the saturation fraction are *completely invariant* to C_{large} across four orders of magnitude (CV(ρ) = 0.000, CV(loss) = 0.005). Since no patch coordinate activates the clipping condition (see above), the value assigned at the pole is never used

in practice; the tanh compression further ensures that even if an extreme value were occasionally assigned, it would be mapped to ± 1 before projection.

Table 3: Sensitivity to the lattice scaling factor $\alpha_u = \alpha_v$.

$\alpha_u = \alpha_v$	ρ	Proxy loss	Mean$\lvert f \rvert$
0.20	0.613	4.464	0.323
0.40	**0.625**	**4.488**	**0.106**
0.60	0.620	4.473	0.053
0.80	0.513	4.469	0.040
1.00	0.347	4.556	0.073
1.20	0.089	4.478	0.100

(C) Lattice scaling factors $\alpha_u = \alpha_v$. The lattice scaling factors are the *only* hyperparameters that meaningfully affect ρ (CV $= 0.418$): values in $\alpha \in [0.2, 0.6]$ produce strong distance decay ($\rho \in [0.61, 0.63]$), whereas larger values push patch coordinates deeper into higher-frequency oscillation regions of $\wp(z)$, reducing the monotonicity of the decay. Importantly, this sensitivity is *by design*: α_u and α_v are the primary geometric control of the encoding, analogous to choosing frequency bands in sinusoidal PE. In all our experiments $\alpha_u = \alpha_v = 0.4$ is set as the default, and the learnable parameter ω_3' (Eq. 2) allows the aspect ratio to adapt during training. The proxy loss varies only within 1% ($\Delta = 0.007$) across the entire range, confirming that training stability is not sensitive to this choice even when ρ degrades at extreme values.

Table 4: Sensitivity to the tanh compression factor α_{scale}.

α_{scale}	ρ	Proxy loss	Sat.%	$\approx$0%
0.010	0.634	4.485	0.26	84.82
0.050	0.630	4.559	0.89	42.98
0.100	0.627	4.434	1.40	20.92
0.150	**0.625**	**4.442**	**1.91**	**11.61**
0.300	0.625	4.492	3.32	5.87
0.500	0.626	4.486	4.34	3.57

(D) tanh **compression factor α_{scale}.** ρ is essentially flat across the entire range (CV $= 0.005$, $\Delta\rho < 0.009$), confirming that the distance-decay property is stable. The proxy loss fluctuates within ± 0.06 (CV $= 0.009$), within the noise of 30 gradient steps. The feature utilisation trade-off is visible: very small α_{scale} wastes representational capacity (84.8% of features near zero at $\alpha = 0.01$), while very large values increase saturation. The default $\alpha_{\text{scale}} = 0.15$ balances these two effects. In practice α_{scale} is implemented as a learnable parameter via `softplus`, so it adapts automatically during training.

Overall conclusion. Table 5 summarises the coefficient of variation (CV) across all sweeps.

Two hyperparameters (κ, C_{large}) are *structurally inert* under standard coordinate normalisation and can be set to any reasonable value without consequence. The tanh compression factor α_{scale} has negligible effect on ρ and loss (CV < 0.01) and is in any case learnable. Only the lattice scaling

Table 5: Coefficient of variation (CV = std/mean) of key metrics across each hyperparameter sweep. CV < 0.01 indicates negligible sensitivity.

Sweep	CV(ρ)	CV(loss)	CV(mean$\lvert f \rvert$)
A: threshold κ	0.0000	0.0059	0.0000
B: C_{large}	0.0000	0.0049	0.0000
C: $\alpha_u = \alpha_v$	0.4175	0.0070	0.8230
D: α_{scale}	0.0049	0.0091	0.6610

factors α_u, α_v require care: the interval $[0.2, 0.6]$ consistently yields $\rho \geq 0.61$, and the learnable ω_3' automatically compensates for aspect-ratio variation within this range during training. These results confirm that WePE is robust and genuinely plug-and-play across all hyperparameters except the spatial frequency range, which is controlled by the single learnable parameter ω_3' and the fixed default $\alpha = 0.4$.